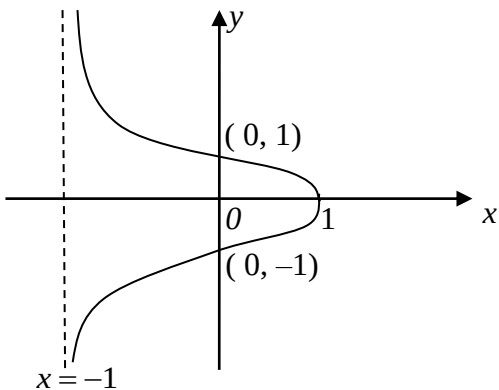
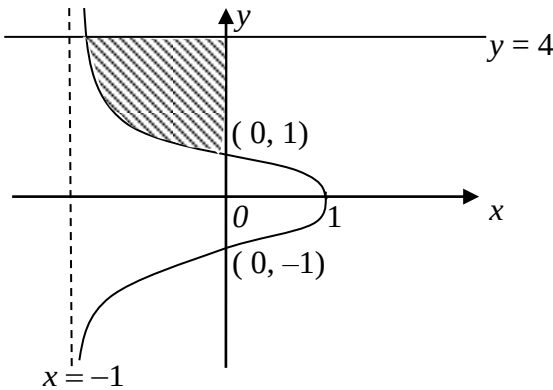


Qn	Suggested Solution
1	$f(x) = ax^3 + bx^2 + cx + d$ where $a, b, c, d \in \mathbb{R}$. Given $f(0) = 1$, $d = 1$. $f'(x) = 3ax^2 + 2bx + c$ From the graph of $y = f'(x)$, $f'(-2) = 0 \Rightarrow 12a - 4b + c = 0 \quad (1)$ $f'(7) = 0 \Rightarrow 147a + 14b + c = 0 \quad (2)$ $f'(2.5) = -9 \Rightarrow 18.75a + 5b + c = -9 \quad (3)$ OR $f''(2.5) = 0 \Rightarrow 15a + 2b = 0 \quad (4)$ Using GC to solve (1), (2) & (3): $a = \frac{4}{27}, b = -\frac{10}{9}, c = -\frac{56}{9}$ $\therefore f(x) = \frac{4x^3}{27} - \frac{10x^2}{9} - \frac{56x}{9} + 1$
	For $f(x)$ is concave downwards $\Leftrightarrow f''(x) < 0$, $x < 2.5$.
2(i)	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 6 \\ 3 \\ 0 \end{pmatrix}$ $\overrightarrow{AP} = \begin{pmatrix} 1+2\lambda \\ -2+\lambda \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2\lambda+2 \\ \lambda+1 \\ 0 \end{pmatrix} = \frac{(\lambda+1)}{3} \begin{pmatrix} 6 \\ 3 \\ 0 \end{pmatrix}$ Since $\overrightarrow{AP} = \frac{\lambda+1}{3} \overrightarrow{AB}$ for $\lambda \neq -1$ and A is a common point, this shows that A, B and P are collinear.

2(ii)	Given that area of triangle OAP is $162\sqrt{5}$, $\frac{1}{2} \vec{OA} \times \vec{AP} = 162\sqrt{5}$ $\frac{1}{2} \left \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix} \times \frac{\lambda+1}{3} \vec{AB} \right = 162\sqrt{5}$ $\frac{1}{2} \left \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix} \times \frac{\lambda+1}{3} \begin{pmatrix} 6 \\ 3 \\ 0 \end{pmatrix} \right = 162\sqrt{5}$ $ \lambda+1 \left \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right = 324\sqrt{5}$ $ \lambda+1 \left \begin{pmatrix} -2 \\ 4 \\ 5 \end{pmatrix} \right = 324\sqrt{5}$ $ \lambda+1 \sqrt{(-2)^2 + (4)^2 + (5)^2} = 324\sqrt{5}$ $ \lambda+1 \sqrt{45} = 324\sqrt{5}$ $ \lambda+1 = 108$ $\lambda+1 = 108 \text{ or } \lambda+1 = -108$ $\lambda = 107 \text{ or } \lambda = -109$ Since P is on BA produced, $\vec{AP} = k\vec{AB}$ for a negative value of k. Hence $\lambda = -109$.
3 1st part	$f(x) = e^{x^2} (\sqrt{1+2x})$ $= (1+x^2+\dots) \left(1 + \frac{1}{2}(2x) + \frac{\frac{1}{2}(-\frac{1}{2})}{2!}(2x)^2 + \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{3!}(2x)^3 + \dots \right)$ $= (1+x^2+\dots) \left(1+x-\frac{1}{2}x^2+\frac{1}{2}x^3+\dots \right)$ $= \left(1+x-\frac{1}{2}x^2+\frac{1}{2}x^3+\dots \right) + (x^2+x^3+\dots)$ $= 1+x+\frac{1}{2}x^2+\frac{3}{2}x^3+\dots$
3(a)	When $x = \frac{1}{3}$,

	$e^{\left(\frac{1}{3}\right)^2} \sqrt{1+2\left(\frac{1}{3}\right)} \approx 1 + \frac{1}{3} + \frac{1}{2}\left(\frac{1}{3}\right)^2 + \frac{3}{2}\left(\frac{1}{3}\right)^3$ $e^{\frac{1}{9}} \sqrt{\frac{5}{3}} \approx 1 + \frac{1}{3} + \frac{1}{2}\left(\frac{1}{3}\right)^2 + \frac{3}{2}\left(\frac{1}{3}\right)^3 = \frac{13}{9}$ $\sqrt{135}e^{\frac{1}{9}} = 9\sqrt{\frac{5}{3}}e^{\frac{1}{9}} \approx 9 \times \frac{13}{9} = 13$
3(b)	$f'(x) = 1 + x + \frac{9}{2}x^2 + \dots$ Using $f'(x) = 1 + x + \frac{9}{2}x^2 + \dots$, $f'(x) = 2xe^{x^2}\sqrt{1+2x} + \frac{e^{x^2}}{\sqrt{1+2x}}$ $f'(x) = 2xf(x) + \frac{e^{x^2}}{\sqrt{1+2x}}$ $\frac{e^{x^2}}{\sqrt{1+2x}} = f'(x) - 2xf(x)$ $= \frac{d}{dx}\left(1 + x + \frac{1}{2}x^2 + \frac{3}{2}x^3 + \dots\right)$ $- 2x\left(1 + x + \frac{1}{2}x^2 + \frac{3}{2}x^3 + \dots\right)$ $= \left(1 + x + \frac{9}{2}x^2 + \dots\right) - (2x + 2x^2 + \dots)$ $= 1 - x + \frac{5}{2}x^2 + \dots$ Alternative method: $\frac{e^{x^2}}{\sqrt{1+2x}} = e^{x^2}(\sqrt{1+2x})(1+2x)^{-1}$ $= \left(1 + x + \frac{1}{2}x^2 + \frac{3}{2}x^3 + \dots\right)(1 - 2x + 4x^2 + \dots)$ $= 1 - 2x + 4x^2 + x - 2x^2 + \frac{1}{2}x^2$ $= 1 - x + \frac{5}{2}x^2 + \dots$

4(i)	$x = ut^2$ $\frac{dx}{dt} = 2tu + t^2 \frac{du}{dt}$ $t^2 \left(2tu + t^2 \frac{du}{dt} \right) - 2t(ut^2) + (ut^2)^2 = 0$ $t^4 \frac{du}{dt} = -u^2 t^4$ $\frac{du}{dt} = -u^2 \text{ (shown)}$
4(ii)	$\int \frac{1}{u^2} du = \int -1 dt$ $-\frac{1}{u} = -t + C'$ $\frac{t^2}{x} = t - C'$ $x = \frac{t^2}{t + C} \quad \text{where } C = -C'$ Since there was 0.2 milligrams of bacteria after 15 minutes, then $0.2 = \frac{(0.25)^2}{0.25 + C} \Rightarrow 0.05 + 0.2C = 0.0625$ $\Rightarrow C = \frac{1}{16}$ $\therefore x = \frac{16t^2}{16t + 1}$
	When $t = 4$, $\therefore x = \frac{16(4)^2}{16(4) + 1} = \frac{256}{65}$ or 3.94
4(iii)	As $t \rightarrow \infty$, $\frac{16t^2}{16t + 1} \rightarrow \infty$. The particular solution of the DE suggests that the <u>amount of bacteria in the Petri dish will grow indefinitely as time passes</u>. Hence the model is not a realistic one.

5(i)	
5(ii)	$x = \cos 2t, \quad y = \tan t$ $\frac{dx}{dt} = -2\sin 2t, \quad \frac{dy}{dt} = \sec^2 t$ $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt} = \frac{\sec^2 t}{-2\sin 2t}$ When $t = \frac{\pi}{3}$, $P(-0.5, \sqrt{3})$ and $\frac{dy}{dx} = \frac{\sec^2 \frac{\pi}{3}}{-2\sin \frac{2\pi}{3}} = -\frac{4}{\sqrt{3}}$ So, $\frac{\sqrt{3}-0}{-0.5-b} = \frac{\sqrt{3}}{4}$ $b = -4.5$
5(iii)	 $x = \cos 2t = 0 \Rightarrow t = \frac{\pi}{4} \Rightarrow y = \tan \frac{\pi}{4} = 1$

	Area $= \left \int_1^4 x dy \right \text{ or } - \int_1^4 x dy$ $= - \int_{\frac{\pi}{4}}^{\tan^{-1} 4} \cos 2t \sec^2 t dt$ $= - \int_{\frac{\pi}{4}}^{\tan^{-1} 4} (2 \cos^2 t - 1) \sec^2 t dt$ $= - \int_{\frac{\pi}{4}}^{\tan^{-1} 4} 2 - \sec^2 t dt$ $= - \left[2t - \tan t \right]_{\frac{\pi}{4}}^{\tan^{-1} 4}$ $= -(2 \tan^{-1} 4 - 4) + \left(\frac{\pi}{2} - 1 \right)$ $= 3 + \frac{\pi}{2} - 2 \tan^{-1} 4$
6	$x = 1$ --- (1) $2x + y + az = 5$ --- (2) Substitute (1) into (2): $2 + y + az = 5$ $y = 3 - az$ Let $z = \lambda$, Hence, $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 3 - a\lambda \\ \lambda \end{pmatrix}$, where λ is a real number. (Shown)
6(a)	The angle between l and p_3 is 60°. This implies $\sin 60^\circ = \frac{\left \begin{pmatrix} 0 \\ -a \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right }{\left(\sqrt{a^2 + 1} \right) \left(\sqrt{1^2 + 2^2 + 1^2} \right)}$ $\Rightarrow \frac{\sqrt{3}}{2} = \frac{ -2a + 1 }{\left(\sqrt{a^2 + 1} \right) (\sqrt{6})}$ $\Rightarrow 4(4a^2 - 4a + 1) = 18(a^2 + 1)$ $\Rightarrow 16a^2 - 16a + 4 = 18a^2 + 18$ $\Rightarrow 2a^2 + 16a + 14 = 0$ $\Rightarrow a^2 + 8a + 7 = 0$ $\Rightarrow (a + 7)(a + 1) = 0$ $\Rightarrow a = -7 \text{ or } a = -1$

6(b)	Since $\overrightarrow{ON}$ is parallel to the normal vector of p_3, $\begin{aligned}\overrightarrow{ON} &= \pm \frac{\sqrt{6}}{3} \left(\frac{1}{\sqrt{1^2 + 2^2 + 1^2}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right) \\ &= \pm \frac{\sqrt{6}}{3} \left(\frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right) \\ &= \begin{pmatrix} 1/3 \\ 2/3 \\ 1/3 \end{pmatrix} \text{ or } \begin{pmatrix} -1/3 \\ -2/3 \\ -1/3 \end{pmatrix}\end{aligned}$ Alternative Method: Let N be the foot of perpendicular from the origin to p_3. Since $\overrightarrow{ON}$ is parallel to the normal vector of p_3, and $ \overrightarrow{ON} = \frac{\sqrt{6}}{3}$ $\sqrt{\alpha^2 + 4\alpha^2 + \alpha^2} = \frac{\sqrt{6}}{3}$ $\sqrt{6\alpha^2} = \frac{\sqrt{6}}{3},$ $6\alpha^2 = \frac{6}{9}$ $\alpha = \pm \frac{1}{3}$ $\overrightarrow{ON} = \begin{pmatrix} 1/3 \\ 2/3 \\ 1/3 \end{pmatrix} \text{ or } \begin{pmatrix} -1/3 \\ -2/3 \\ -1/3 \end{pmatrix}$
6(c)	Given that p_1, p_2 and p_3 do not have common point, then line l must be parallel to p_3. Hence $\begin{pmatrix} 0 \\ -a \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 0$

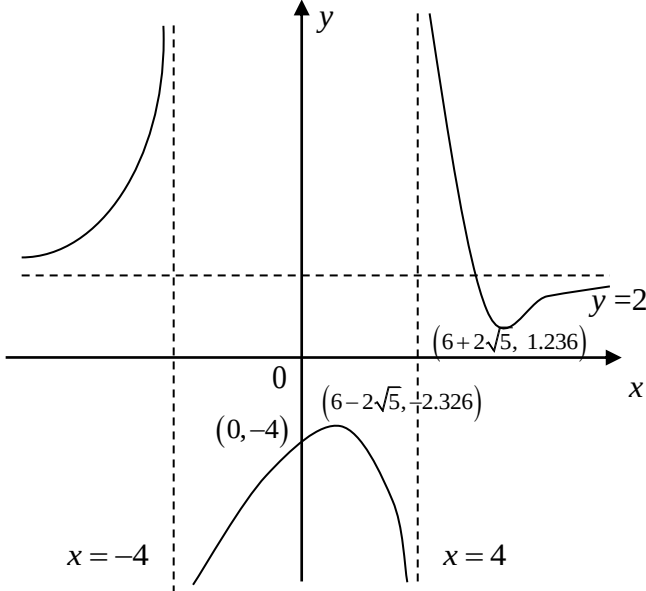
	$-2a + 1 = 0$ $a = \frac{1}{2}$ Also, a point $(1, 3, 0)$ in l must not lie in p_3 . Hence $\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \neq b$ $b \neq 7$
7 (a)(i)	$r = \frac{a+d}{a+3d} = \frac{a+3d}{a+8d}$ $\Rightarrow (a+d)(a+8d) = (a+3d)^2$ $\Rightarrow a^2 + 9ad + 8d^2 = a^2 + 6ad + 9d^2$ $\Rightarrow d^2 - 3ad = 0$ $\Rightarrow d(d - 3a) = 0$ $\Rightarrow d = 0 \text{ (rej.) or } d = 3a \text{ (shown)}$ Alternatively, let b be the first term of the geometric series. Then $d = \frac{b-br}{5} = \frac{br-br^2}{2}$ $\Rightarrow 2b - 2br = 5br - 5br^2$ $\Rightarrow 5r^2 - 7r + 2 = 0$ $\Rightarrow (5r-2)(r-1) = 0$ $\Rightarrow r = \frac{2}{5} \text{ or } r = 1 \text{ (rej because otherwise } d = 0)$ Hence $d = \frac{b - b\left(\frac{2}{5}\right)}{5} = -\frac{3}{25}b = -\frac{3}{25}(a+8d)$ $25d = -3a - 24d$ $d = 3a \text{ (shown)}$
7 (a) (ii)	$r = \frac{a+d}{a+3d} = \frac{a+3a}{a+9a} = \frac{4a}{10a} = \frac{2}{5}.$ Since $\left \frac{2}{5}\right < 1$, the geometric series is convergent.

	$\begin{aligned}\text{Sum to infinity} &= \frac{a+8d}{1-r} \\ &= \frac{a+24a}{1-\frac{2}{5}} \\ &= \frac{5}{3}(25a) \\ &= \frac{125}{3}a\end{aligned}$
7(b) (i)	The distance the mountaineer climbs for each hour follows an arithmetic progression with first term 300 metres and common difference (-10) metres. Total distance travelled after n hours $\leq x$ $\frac{n}{2}[2(300) + (n-1)(-10)] \leq x$ $\frac{n}{2}(600 - 10n + 10) \leq x$ $\frac{n}{2}(610 - 10n) \leq x$ $n(305 - 5n) \leq x$ $-5n^2 + 305n \leq x \text{ (shown)}$ $p = -5, q = 305$
7(b) (ii)	If $x = 2500$, then $-5n^2 + 305n \leq 2500$ $-5n^2 + 305n - 2500 \leq 0$ $n \leq 9.757 \text{ or } n \geq 51.24$ Hence $n = 9$.
8	$p(-3) = 0$ $\Rightarrow (-3)^3 + m(-3)^2 - 7(-3) + 15 = 0$ $\Rightarrow -27 + 9m + 21 + 15 = 0$ $\therefore m = -1$
8(i)	$z^3 - z^2 - 7z + 15 = 0$ $(z+3)(z^2 - 4z + 5) = 0$ $z = -3 \text{ or } z = \frac{-(-4) \pm \sqrt{16 - 4(1)(5)}}{2(1)}$ $= \frac{4 \pm \sqrt{-4}}{2}$ $= 2 \pm i$ $\therefore z_1 = 2 - i, z_2 = 2 + i, z_3 = -3$

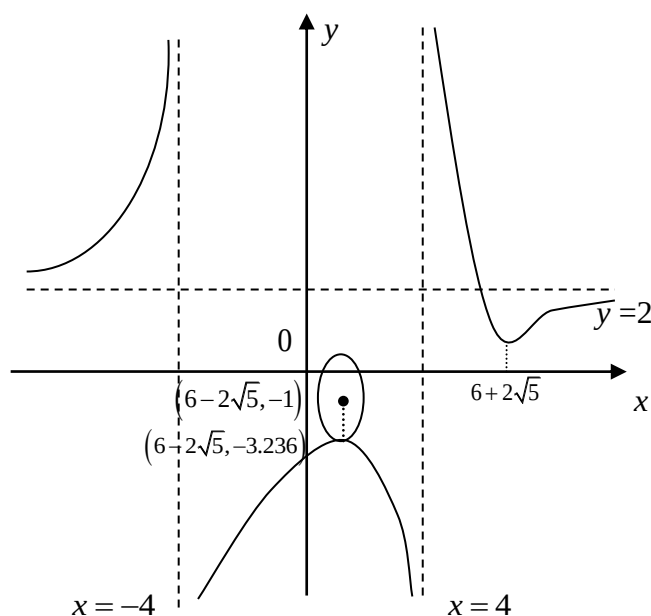
8(ii)	Since z_1 and z_2 are complex conjugates, $(z_1^n)^* = (z_1^*)^n = z_2^n$. z_1^n and z_2^n are complex conjugates as well. Thus $z_1^n - z_2^n = z_1^n - (z_1^n)^*$ $= (2i) \operatorname{Im}(z_1^n)$ which is purely imaginary. Alternative method: $z_1^n - z_2^n = (2-i)^n - (2+i)^n$ $= \left(\sqrt{5} e^{-i(\tan^{-1} 0.5)} \right)^n - \left(\sqrt{5} e^{i(\tan^{-1} 0.5)} \right)^n$ $= 5^{\frac{n}{2}} \left(e^{-i(\tan^{-1} 0.5)n} - e^{i(\tan^{-1} 0.5)n} \right)$ $= 5^{\frac{n}{2}} \begin{pmatrix} \cos(n \tan^{-1} 0.5) - i \sin(n \tan^{-1} 0.5) \\ -\cos(n \tan^{-1} 0.5) - i \sin(n \tan^{-1} 0.5) \end{pmatrix}$ $= -(2) 5^{\frac{n}{2}} i \sin(n \tan^{-1} 0.5)$ Since $\operatorname{Re}(z_1^n - z_2^n) = 0$, $z_1^n - z_2^n$ is purely imaginary.
8(iii)	Since $z_1 = \sqrt{5}$, $z_3 = 3$, $z_1 \neq z_3$. The locus of complex numbers satisfying the equation $w = a$, for some positive constant a, will not pass through all the points representing the complex numbers z_1, z_2 and z_3.
9(i)	Let P_n be the statement $u_n = \frac{1}{n(n+1)}$, where $n \in \mathbb{Z}^+$. When $n = 1$, $LHS = u_1 = \frac{1}{2} \text{ (Given)}$ $RHS = \frac{1}{(1)(1+1)} = \frac{1}{2}$ Hence, P_1 is true. Suppose/assume P_k is true for some $k \in \mathbb{Z}^+$, i.e. $u_k = \frac{1}{k(k+1)}$.

	Need to show that P_{k+1} is also true, i.e. $u_{k+1} = \frac{1}{(k+1)(k+2)}$ $ \begin{aligned} u_{k+1} &= u_k - \frac{2}{k(k+1)(k+2)} \\ &= \frac{1}{k(k+1)} - \frac{2}{k(k+1)(k+2)} \\ &= \frac{(k+2)-2}{k(k+1)(k+2)} \\ &= \frac{k}{k(k+1)(k+2)} \\ &= \frac{1}{(k+1)(k+2)} \end{aligned} $ Hence P_{k+1} is true if P_k is true. Since P_1 is true and P_k is true implies that P_{k+1} is true, by mathematical induction, $u_n = \frac{1}{n(n+1)}$ for all positive integers n.
9(ii)	$ \begin{aligned} \sum_{r=1}^N \frac{1}{r(r+1)(r+2)} &= \frac{1}{2} \sum_{r=1}^N (u_r - u_{r+1}) \\ &= \frac{1}{2} (u_1 - u_2 \\ &\quad + u_2 - u_3 \\ &\quad + u_3 - u_4 \\ &\quad \vdots \\ &\quad + u_N - u_{N+1}) \\ &= \frac{1}{2} (u_1 - u_{N+1}) \\ &= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{(N+1)(N+2)} \right) \end{aligned} $
9(iii)	$ \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} \dots = \lim_{N \rightarrow \infty} \sum_{r=1}^N \frac{1}{(r+2)^3} $ For $r \in \mathbf{Z}^+$, $ (r+2)^3 > r(r+1)(r+2) \text{ OR } $ $ \frac{1}{(r+2)^3} < \frac{1}{r(r+1)(r+2)} \text{ OR } $ $ \sum_{r=1}^N \frac{1}{(r+2)^3} < \sum_{r=1}^N \frac{1}{r(r+1)(r+2)} = \frac{1}{4} - \frac{1}{2(N+1)(N+2)} < \frac{1}{4} $ Hence $ \lim_{N \rightarrow \infty} \sum_{r=1}^N \frac{1}{(r+2)^3} < \lim_{n \rightarrow \infty} \frac{1}{4} = \frac{1}{4} $

9(iv)	Let $r - 2 = j$. Then $r - 1 = j + 1$ and $r = j + 2$. $\sum_{r=10}^N \frac{1}{r(r-1)(r-2)} = \sum_{j+2=10}^{j+2=N} \frac{1}{j(j+1)(j+2)}$ $= \sum_{j=8}^{j=N-2} \frac{1}{j(j+1)(j+2)}$ $= \sum_{j=1}^{j=N-2} \frac{1}{j(j+1)(j+2)} - \sum_{j=1}^{j=7} \frac{1}{j(j+1)(j+2)}$ $= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{N(N-1)} \right) - \frac{1}{2} \left(\frac{1}{2} - \frac{1}{(8)(9)} \right)$ $= \frac{1}{144} - \frac{1}{2N(N-1)}$
10 (a)(i)	$y = \frac{2x^2 - a^2x + 4a^2}{x^2 - a^2}$ By long division, $y = 2 + \frac{6a^2 - a^2x}{x^2 - a^2}$ Vertical asymptotes: $x = -a$ or $x = a$ Horizontal asymptote: $y = 2$
10 (a) (ii)	From $y = 2 + \frac{6a^2 - a^2x}{x^2 - a^2}$, $\frac{dy}{dx} = 0 + \frac{(-a^2)(x^2 - a^2) - (6a^2 - a^2x)(2x)}{(x^2 - a^2)^2}$ $= \frac{(a^2)(-x^2 + a^2 - 12x + 2x^2)}{(x^2 - a^2)^2}$ $= \frac{(a^2)(2x^2 - 12x + a^2)}{(x^2 - a^2)^2}$

	$\frac{dy}{dx} = 0$ $\Rightarrow \frac{(a^2)(x^2 - 12x + a^2)}{(x^2 - a^2)^2} = 0$ $\Rightarrow x^2 - 12x + a^2 = 0$ Given C has two turning points, $\therefore b^2 - 4ac > 0$ $\Rightarrow (-12)^2 - 4(1)(a^2) > 0$ $144 - 4a^2 > 0$ $a^2 - 36 < 0$ $(a - 6)(a + 6) < 0$ $-6 < a < 6$ Since a is a positive constant, $0 < a < 6$. (Shown)
10	
	$h^2(x - 6 + 2\sqrt{5})^2 + (y + 1)^2 = h^2$ $\Rightarrow (x - 6 + 2\sqrt{5})^2 + \frac{(y + 1)^2}{h^2} = 1$ It is an ellipse centred at $(6 - 2\sqrt{5}, -1)$.

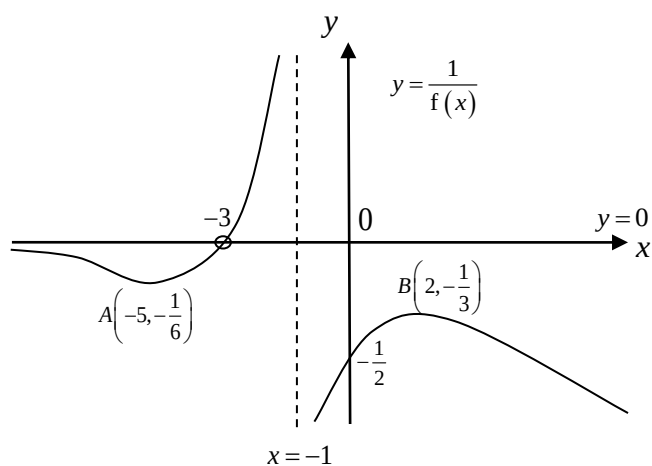
Maximum turning point of C occurs at $(6 - 2\sqrt{5}, -3.236)$.

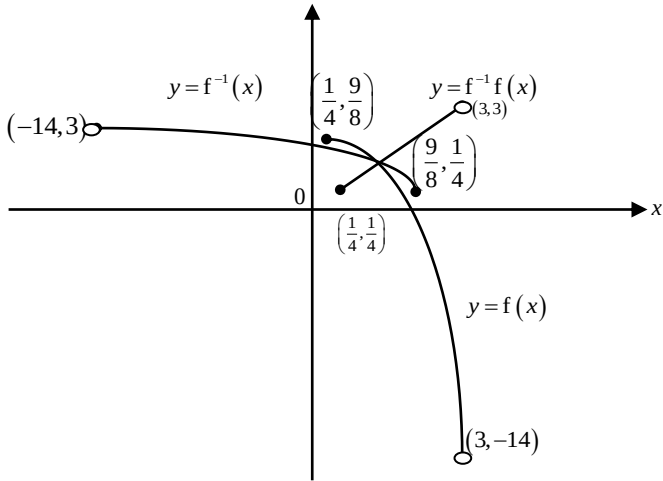


For the ellipse to intersect the curve C more than once, $h > 2.236..$

Since h is a positive integer, $h \geq 3$.

10
(b)



11(i)	
	$f^2(x) = x$ $f(x) = f^{-1}(x)$ $f(x) = x$ $-2x^2 + x + 1 = x$ $2x^2 = 1$ $x = \frac{\sqrt{2}}{2} \text{ or } -\frac{\sqrt{2}}{2}$ Since $x \geq \frac{1}{4}$, $x = \frac{\sqrt{2}}{2}$.
11(ii)	$R_g = (-\ln 2, \ln(1-2k)) \subseteq (-14, 9/8] = D_{f^{-1}}$ For least k, $\ln(1-2k) = 9/8$ $k = (1 - e^{9/8})/2 \text{ or } -1.03$
11(iii)	$f(x) \leq h(x)$ $x - 2x^2 + 1 \leq 18 x - 2 - 18$ Consider $x \geq 2$ $x - 2x^2 + 1 \leq 18(x - 2) - 18$ $2x^2 + 17x - 55 \geq 0$ $(2x - 5)(x + 11) \geq 0$ $x \leq -11 \text{ or } x \geq 2.5$ Then $x \geq 2.5$

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	Consider $x \leq 2$ $x - 2x^2 + 1 \leq 18(-x + 2) - 18$ $2x^2 - 19x + 17 \geq 0$ $(x - 8.5)(x - 1) \geq 0$ $x \leq 1 \text{ or } x \geq 8.5$ Then $x \leq 1$ $\frac{1}{4} \leq x \leq 1 \text{ or } 2.5 \leq x < 3$
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